This literature review synthesizes the provided sources to explore the intersection of **sustainable artificial intelligence, energy efficiency, data governance, and multilingual NLP performance**.

Project Concept/Overview

The core concept emerging from these sources is the transition toward **Sustainable AI**, which prioritizes the energy efficiency of AI models alongside their predictive accuracy(Budennyy et al. 2022) . Current deep learning (DL) models are growing exponentially in complexity, with some reaching over a trillion parameters, leading to a massive increase in energy consumption during training and inference(Budennyy et al. 2022) . This necessitates tools like **eco2AI**, which track equivalent CO_2 emissions to encourage the development of optimal, low-cost architectures(Budennyy et al. 2022) . Parallel to environmental sustainability, there is a focus on **Sovereign AI**, where communities retain ownership of data and AI outputs while building local infrastructure to cut foreign dependency.

Pillar Identification

The research identifies four primary pillars for a responsible and localized AI infrastructure:

1. **AI Sovereignty**: African communities retaining ownership of data and systems.
2. **Local AI Infrastructure**: Keeping computing and hosting within South Africa.
3. **Domestic AI Models**: Training models on local languages and lived experiences.
4. **Community Impact**: Bridging the gap between academic research and grounded African needs.

Additionally, the **Sustainable AI path** focuses on optimizing the technology itself, while the **Green AI path** uses AI to achieve external sustainability goals, such as smart grid design and climate modeling(Budennyy et al. 2022) .

Problem Identification/Research

The sources highlight three critical problem areas:

- **Environmental Impact**: AI's indirect carbon footprint is significant; for example, training BERT can emit as much CO_2 (Budennyy et al. 2022) as a transcontinental flight.
- **Language Gaps**: Large Language Models (LLMs) perform unevenly across African languages due to a lack of digital representation, as evidenced by the **AFROBENCH** evaluation of 64 African languages(Shikwambane 2026) .
- **Governance Uncertainties**: Implementing the **Protection of Personal Information Act (POPIA)** for cross-border data sharing is legally complex and often vague, particularly regarding "adequacy" requirements for recipient countries(Abdulrauf, Adaji & Ojibara 2024) .

Governance System

Governance is primarily structured around **POPIA (Act 4 of 2013)**, which regulates the free flow of information while protecting privacy(Abdulrauf, Adaji & Ojibara 2024) (Coetzee 2024) . A **POPIA Compliance Framework** has been developed by the Academy of Science of South Africa (ASSAf) to serve as industry best practice for researchers. Furthermore, **Section 72 of POPIA** prescribes specific conditions for transferring data to third parties in foreign countries, requiring either an "adequate level of protection" in the recipient country, data subject consent, or contractual necessity(Coetzee 2024) (Abdulrauf, Adaji & Ojibara 2024) .

Related Work Comparison Table

Based on the methodologies for energy monitoring and AI tracking in the sources:

Feature	eco2AI (Budennyy et al. 2022)	XDL-Energy (M.Sadeeq 2025)	MNR Model (Tavakkoli, Lai, Chen, Fang & Lee 2025)	SAEEC Method (Luna-Romero, Serrano-Guerrero, de Souza & Escrivá-Escrivà 2024)
Primary Goal	CO_2 tracking for ML models	Smart campus forecasting	Sensorless PC power measurement	Building anomaly detection
Technique	Python library, IP-based emission coeffs	Hybrid CNN-LSTM + SHAP	Multivariate Nonlinear Regression	Statistical analysis (Z-values)
Hardware Focus	CPU, GPU, RAM	IoT sensor networks	PC (CPU/GPU) via magnetometers	Building-wide electrical load
Interpretability	Regional emission database	Global/Local SHAP values	Semi-empirical coefficients	Multi-criteria cataloging

Scenario

A common application scenario is the **Smart Campus**, where distributed IoT sensors collect real-time data on energy use, temperature, and occupancy(M.Sadeeq 2025) . In this environment, unexpected spikes in energy—such as an HVAC system running in an empty lab during non-working hours—can be flagged as anomalies(M.Sadeeq 2025) . Another scenario involves **multilingual contact centers**, where resource-efficient AI like **Vulavula** is used for real-time speech-to-text and translation in environments with limited connectivity("Vulavula — Multilingual Speech-to-Text & Translation | Lelapa AI" n.d.) .

Methodology

The reviewed research employs several advanced methodologies:

- **Hybrid Deep Learning**: Using **CNN-LSTM** architectures to capture both spatial patterns in multi-sensor data and temporal dependencies in usage sequences(M.Sadeeq 2025) .
- **Explainable AI (XAI)**: Integrating **SHAP (SHapley Additive exPlanations)** to provide feature-level transparency, showing stakeholders *why* a model predicted a certain energy load or flagged an anomaly(M.Sadeeq 2025) .
- **Sensorless Measurement**: Utilizing **CMOS MEMS magnetometers** to measure the magnetic field around a PC's power wire, allowing for accurate power estimation without traditional hardware sensors(Tavakkoli, Lai, Chen, Fang & Lee 2025) .
- **Statistical Normalization**: Converting consumption data to **Z-scores** to identify data points that deviate from established patterns(Luna-Romero, Serrano-Guerrero, de Souza & Escrivá-Escrivà 2024) .

Findings on Energy Wastage Detection

Research into energy wastage provides the following findings:

- **Idle Computer Waste**: A study at Dalhousie University found that **71% of campus computers were left on while unused**, contributing to significant avoidable emissions and costs(Anderson et al. 2005) .
- **Misconceptions**: Many users wrongly believe that screensavers save energy or that turning computers off shortens their lifespan; in reality, **turning them off benefits their lifespan** and reduces mechanical stress(Anderson et al. 2005) .
- **Detection RI (Relative Improvement)**: Using computational intelligence for anomaly detection offered a **12.41% improvement in accuracy** and a **42% reduction in false rates** compared to purely statistical methods(Luna-Romero, Serrano-Guerrero, de Souza & Escrivá-Escrivà 2024) .
- **Actionable Insights**: Anomaly detection frameworks can pinpoint specific drivers of waste, such as **temperature thresholds (~34°C)** that trigger inefficient HVAC transitions, allowing managers to implement schedule-aware setback(M.Sadeeq 2025)

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